

AI-Driven Analysis of OPD Appointment No-Shows

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Abstract

Outpatient department (OPD) appointment no-shows generate significant operational and financial burdens for healthcare systems globally, with estimated revenue losses of 3–14% per facility annually and systemic capacity waste that delays care for waiting patients. This capstone project applies artificial intelligence and machine learning techniques to 110,522 real OPD appointment records from Brazilian public hospitals (the Kaggle Medical Appointment No-Shows dataset) to address four hypothesis-driven business questions examining patient risk profiling, scheduling lead-time effects, SMS reminder moderation, and geographic resource utilization patterns. Problem: OPD no-show rates of 15–30% cause idle clinical capacity, revenue leakage of 3–14% annually per facility, and delayed care for waiting patients. Solution Approach: The analytical pipeline employs Logistic Regression, Random Forest, XGBoost with SHAP (SHapley Additive exPlanations) explainability, interaction-term regression, and K-Means clustering. Data: Kaggle Medical Appointment No-Shows dataset (Arroba, 2016); N = 110,522 appointment records; unit of analysis = individual OPD appointment; timeframe = November 2015 – June 2016, Vitória, Brazil. Technology: Python 3.x; Pandas, Scikit-learn, XGBoost, SHAP, Fairlearn, Matplotlib, Seaborn; CPU-based local compute (Jupyter Notebook). Major Results: Tuned XGBoost achieved AUC-ROC = 0.727 and Sensitivity = 90.4% on held-out test data, versus Logistic Regression baseline AUC = 0.586. The model is the only evaluated classifier satisfying the clinical fairness threshold (Demographic Parity Difference = +0.034 < 0.05). Implementation Area: The model is designed for integration into hospital OPD scheduling systems as a real-time risk scoring engine, serving hospital operations managers, revenue cycle teams, patient engagement coordinators, and public health planners.

Keywords: OPD appointment no-shows, machine learning, XGBoost, SHAP, wait-time gap, SMS effectiveness, health equity, hospital revenue optimization, K-Means clustering

Introduction

Background and Context

Outpatient department (OPD) appointment no-shows are instances where a scheduled patient fails to attend without prior cancellation. These represent one of the most operationally damaging and financially costly inefficiencies in modern healthcare delivery. Across healthcare systems globally, no-show rates typically range from 15% to 30%, translating into idle clinical staff, wasted infrastructure capacity, delayed care for waiting patients, and estimated revenue losses of 3–14% per facility annually (Al-Surimi et al., 2025). In the United States alone, appointment no-shows are estimated to cost the healthcare system approximately \$150 billion annually (Nair & Bhanu, 2025).

The Kaggle Medical Appointment No-Shows dataset, comprising 110,527 real OPD appointment records from Vitória, Espírito Santo, Brazil, provides a nationally representative and operationally realistic basis for this analysis. With a no-show rate of approximately 20.2% — representing over 22,000 missed appointments — the dataset offers sufficient statistical power for robust hypothesis testing and subgroup-level analysis. Despite this scale, most healthcare facilities continue to rely on reactive, undifferentiated management approaches: blanket SMS reminder campaigns, manual overbooking heuristics, and post-hoc rate reporting without predictive insight.

Machine learning (ML) offers a fundamentally superior alternative. By identifying patterns across large, diverse appointment datasets, ML models can generate individualized no-show risk scores at the point of scheduling, enabling proactive and cost-efficient interventions: priority outreach for high-risk patients, waitlist filling for at-risk slots, and dynamic overbooking calibrated to predicted no-show probability. The present study applies this paradigm to four hypothesis-driven business questions spanning the predict-diagnose-intervene-allocate strategic framework for OPD no-show management.

Problem Statement

The objective of this study is to predict and explain OPD appointment no-show probability (Y = binary no-show outcome) for individual appointment records from Brazilian public hospitals using patient demographics, chronic condition indicators, socioeconomic status, scheduling lead time, and SMS reminder receipt (X = key independent variables) over the period November 2015 – June 2016. Success will be evaluated using AUC-ROC, Sensitivity, F1-Score, and Demographic Parity Difference fairness metrics.

Purpose of the Study

This study aims to accomplish four interrelated objectives:

1. predict individual no-show risk at the point of scheduling using machine learning.
2. diagnose the structural determinants of elevated no-show risk, including lead-time thresholds, patient demographics, and geographic patterns.
3. evaluate the differential effectiveness of SMS reminders across patient subgroups; and
- (4) segment neighbourhoods into actionable geographic risk clusters for targeted resource allocation. Collectively, these objectives support a shift from reactive to proactive, data-driven OPD appointment management.

Research Problems / Research Questions

RQ1: Which combination of patient demographics (age, gender), chronic conditions (hypertension, diabetes, alcoholism, handicap), and socioeconomic status (Scholarship) most significantly predicts OPD appointment no-show probability?

RQ2: At what scheduling lead-time threshold (WaitDays) does no-show probability increase significantly, and does the relationship follow a monotonic pattern?

RQ3: Is the effectiveness of SMS reminders on no-show rates moderated by WaitDays and patient socioeconomic status (Scholarship), and for which subgroups is SMS effective versus counterproductive?

RQ4: Do neighbourhood-level no-show rates differ significantly after controlling for patient demographics and chronic conditions, and can neighbourhoods be clustered into actionable geographic risk segments?

Contributions and Expected Value

Practical Contribution: A deployable XGBoost model with real-time risk scoring, enabling a four-tier intervention protocol calibrated to individual no-show probability.

Technical Contribution: End-to-end pipeline integrating SMOTE-balanced ensemble modeling, SHAP explainability, interaction-term regression, and Fairlearn fairness evaluation on a benchmark healthcare dataset.

Value to Stakeholders: Hospital operations managers, revenue cycle teams, patient engagement coordinators, and public health planners each receive actionable, evidence-based recommendations tied directly to the model outputs.

Healthcare systems implementing predictive no-show management stand to recover a substantial proportion of revenue currently lost to missed appointments. For a facility processing 100 OPD appointments daily at an average consultation cost of \$50, a 20% no-show rate translates to approximately \$250,000 in annual revenue leakage. A five-percentage-point reduction through predictive intervention would recover approximately \$62,500 per facility per year — a return that far exceeds the cost of deploying a machine learning scheduling system.

Literature Review

Literature Review Approach

Sources were selected via targeted searches across PubMed, Google Scholar, IEEE Xplore, and JSTOR using keywords including “outpatient appointment no-show prediction,” “machine learning healthcare scheduling,” “SMS reminder effectiveness,” and “geographic health equity.” Inclusion criteria required: (1) empirical or systematic review studies, (2) publication between 2014 and 2025, and (3) direct methodological or contextual relevance to at least one of the four research questions. The 12 selected sources span four thematic clusters aligned with the study’s RQs.

Summary of Key Literature

Table 1 Literature Relevance Matrix

Author (Year)	Domain/Context	Method	Key Finding	RQ Relevance
Kheirkhah et al. (2016)	Prevalence & predictors	Multi-site analysis	Age, SES, lead time are top predictors	RQ1
Nair & Bhanu (2025)	Financial impact & ML review	Systematic review (47 studies)	ML reduces no-shows by 18–23%	All RQs
Al-Surimi et al. (2025)	AI deployment	Before-after study	AI reduced no-shows in UAE primary care	All RQs
Salazar et al. (2022)	ML techniques	Systematic lit. review	Ensemble models outperform single classifiers	RQ1, RQ4
Hamdan et al. (2023)	ML prediction	RF & GBM	AUC 0.78–0.81; lead time strongest predictor	RQ4
Deina et al. (2024)	Decision analysis	ML + optimization	18% reduction in idle slots	RQ2, RQ4

Author (Year)	Domain/Context	Method	Key Finding	RQ Relevance
Denney et al. (2019)	ML in primary care	LR vs. XGBoost	XGBoost outperforms LR (AUC 0.81 vs. 0.73)	RQ4
Valenzuela- Núñez et al. (2024)	Intelligent scheduling	Optimization + ML	Risk-based overbooking improves efficiency	RQ1, RQ2
Huang & Hanauer (2014)	Lead-time effect	VHA data analysis	Long lead times increase no- shows	RQ2
Valero-Bover et al. (2022)	Reminder targeting	Prospective validation	Targeted reminders reduce no- shows	RQ3
Goffman et al. (2017)	Geographic variation	Multi-site analysis	Geography affects attendance independently	RQ4
Arroba (2016)	Dataset source	Observational dataset	Benchmark dataset for ML research	All RQs

Note. RF = Random Forest; GBM = Gradient Boosting Machine; LR = Logistic

Regression; VHA = Veterans Health Administration; SES = Socioeconomic Status; AUC = Area Under the Curve; ML = Machine Learning; RQ = Research Question.

Prevalence and Financial Impact of Appointment No-Shows

Kheirkhah et al. (2016) conducted a comprehensive multi-site analysis of no-show prevalence across U.S. healthcare settings, documenting rates ranging from 5% to 39% depending on specialty, patient population, and appointment type. The study identified low socioeconomic status, younger patient age, and long scheduling lead times as the strongest demographic predictors. This source directly informs RQ1, enabling comparison of predictor patterns across different healthcare system contexts. Nair and Bhanu (2025) published a systematic review estimating U.S. healthcare losses at approximately \$150 billion annually

due to no-shows and identified the absence of individual-level predictive risk stratification as the primary gap. Their review of 47 ML studies found that predictive models reduced no-show rates by an average of 18–23% in deployment studies, establishing the financial and operational business case for this capstone. Al-Surimi et al. (2025) reported on a real-world AI implementation across Emirates Health Services primary care centres, documenting a 21% baseline no-show rate that was substantially reduced following deployment of an AI-driven scheduling system, providing direct empirical evidence that AI-based interventions generate measurable operational improvements.

Machine Learning Approaches for No-Show Prediction

Salazar et al. (2022) conducted a systematic literature review of ML techniques applied to medical appointment no-shows across 31 studies, finding that ensemble methods — particularly Random Forest and Gradient Boosting — consistently outperformed single classifiers and that feature engineering of wait-time gap, prior no-show history, and appointment day-of-week was critical. Hamdan et al. (2023) applied Random Forest and Gradient Boosting to a Malaysian tertiary hospital, achieving AUC-ROC values of 0.78 and 0.81 respectively, serving as external comparison benchmarks for this study. Deina et al. (2024) proposed a decision analysis framework combining ML prediction with scheduling optimization in a Brazilian hospital system, demonstrating an 18% reduction in idle slot rates through integration of no-show probability scores into overbooking decisions. Denney et al. (2019) found that XGBoost (AUC 0.81) significantly outperformed Logistic Regression (AUC 0.73) in a U.S. primary care setting, directly motivating the choice of XGBoost as the primary model in this study. Valenzuela-Núñez et al. (2024) developed a mathematical framework integrating ML-predicted no-show probabilities into scheduling optimization, providing the operational blueprint for translating risk scores into scheduling recommendations.

Scheduling Lead Time and Reminder Effectiveness

Huang and Hanauer (2014) used Veterans Health Administration data from 380,000 appointments to demonstrate that appointments scheduled more than seven days in advance had significantly higher no-show rates across all specialties, with near-zero rates for same-day appointments. This directly motivates RQ2's WaitDays threshold analysis and the 15-day scheduling policy recommendation. Valero-Bover et al. (2022) conducted a prospective validation finding that targeted reminders sent to high-risk patients reduced no-show rates from 19.3% to 15.9%, whereas uniform reminders produced no significant improvement, providing the primary empirical support for RQ3's moderation hypothesis.

Social Determinants, Geographic Patterns, and Health Equity

Goffman et al. (2017) studied the VHA across 22 facilities and found significant facility-level and geographic heterogeneity in no-show rates not fully explained by patient demographics, with geographic proximity negatively associated with no-show rates across all demographic subgroups. This motivates RQ4, which tests whether neighbourhood-level patterns in the Kaggle dataset contain predictive information beyond patient-level demographics alone. The dataset source (Arroba, 2016) has been referenced in more than 50 peer-reviewed publications, establishing it as the accepted benchmark for healthcare operations ML research.

Materials and Method

Data Source and Description

The primary dataset is the Kaggle Medical Appointment No-Shows dataset (Arroba, 2016), comprising 110,527 real OPD appointment records from public hospitals in Vitória, Espírito Santo, Brazil, collected between November 2015 and June 2016. The dataset is freely available at <https://www.kaggle.com/datasets/joniarroba/noshowappointments> and contains 14 original variables covering appointment scheduling metadata, patient demographics, chronic condition indicators, socioeconomic status (Bolsa Família enrollment), and SMS reminder delivery records. One derived feature, WaitDays, is engineered as the primary operational variable for RQ2. The dataset has no missing values after removing five records with biologically implausible negative ages, yielding a final analytical sample of $n = 110,522$. The class distribution reflects a moderate imbalance: approximately 79.81% attended (Show) and 20.19% missed (No-Show), which informed the application of SMOTE during model training.

Data Dictionary

Table 2 Data Dictionary – Kaggle Medical Appointment No-Shows Dataset

Variable Name	Type	Description / Values	Used In
PatientId	Identifier	Unique patient identifier — not used in modeling	—
AppointmentID	Identifier	Unique appointment identifier — not used in modeling	—
Gender	Binary	Patient sex: F (Female) / M (Male)	RQ1
ScheduledDay	DateTime	Date and time the appointment was booked	RQ2, RQ3
AppointmentDay	DateTime	Date of the actual OPD appointment	RQ2, RQ3
WaitDays*	Continuous	Engineered: AppointmentDay – ScheduledDay (days)	RQ2, RQ3
Age	Continuous	Patient age in years (range: 0–115; 5 negative values removed)	RQ1, RQ3

Variable Name	Type	Description / Values	Used In
Neighbourhood	Categorical	Patient's neighbourhood (81 unique values)	RQ4
Scholarship	Binary	Bolsa Família social welfare enrollment: 1 = Yes, 0 = No	RQ1, RQ3, RQ4
Hipertension	Binary	Diagnosed hypertension: 1 = Yes, 0 = No	RQ1, RQ4
Diabetes	Binary	Diagnosed diabetes: 1 = Yes, 0 = No	RQ1, RQ4
Alcoholism	Binary	Diagnosed alcoholism: 1 = Yes, 0 = No	RQ1
Handicap	Ordinal	Disability level: 0 = None, 1–4 = Degree of handicap	RQ1
SMS_received	Binary	SMS reminder sent: 1 = Yes (≥ 1 message), 0 = No	RQ3
No-show	Binary	TARGET: 'Yes' = did not attend; 'No' = attended	All RQs

Note. WaitDays (*) is an engineered feature derived from ScheduledDay and AppointmentDay.

Table 3 Dataset Profile Summary

Attribute	Detail
Total Records	110,527 appointments (110,522 after removing 5 invalid age records)
Date Range	Scheduling: November 2015 – June 2016 Appointments: April – June 2016
No-Show Rate	~20.2% (22,319 no-shows out of 110,522 appointments)
Show Rate	~79.8% (88,203 shows) — moderate class imbalance
Unique Patients	62,299 distinct PatientIds (some patients appear multiple times)
Neighbourhoods	81 unique neighbourhoods across Vitória, Espírito Santo, Brazil
Gender Distribution	Female: 64.9% Male: 35.1%
Age Range	0–115 years; median ~37 years
Scholarship Recipients	~9.8% of appointments involve Bolsa Família recipients
SMS Reminders Sent	~32.1% of appointments received at least one SMS reminder
Chronic Conditions	Hypertension: 19.7% Diabetes: 7.2% Alcoholism: 3.0% Handicap: 2.0%

Attribute	Detail
Missing Values	None. Dataset is complete after age-filter correction.

Note. Data derived from the Kaggle Medical Appointment No-Shows dataset (Arroba, 2016). N = 110,527 appointments from Brazilian public hospitals, 2016.

Inclusion and Exclusion Criteria

All 110,527 records were initially included. Five records were excluded due to biologically implausible negative age values identified as data entry errors. Records with Age = 115 were retained as biologically plausible extreme values. No additional exclusion criteria were applied. The dataset contains no missing values, and no records were excluded on the basis of missing data. Final analytical sample: n = 110,522.

Data Cleaning and Preprocessing

1. Remove Invalid Records: Five records with Age < 0 removed as data entry errors. Final sample: n = 110,522.
2. Feature Engineering (WaitDays): WaitDays = (AppointmentDay – ScheduledDay).dt.days. Negative values clipped to 0.
3. Recode Target Variable: No-show ‘Yes’ → 1 (no-show), ‘No’ → 0 (show) to align with standard ML convention.
4. Encode Categorical Features: Gender binary-encoded (F = 0, M = 1); Neighbourhood label-encoded (81 levels); Handcap retained as ordinal integer (0–4).
5. WaitDays Bucketing: WaitBucket categorical variable: 0 days, 1–7 days, 8–15 days, 16–30 days, 31+ days.
6. Train/Test Split and Class Imbalance Correction: 80/20 stratified split preserving the 20.19% positive class ratio. SMOTE applied to training set only to prevent data leakage.

Research Hypotheses

The following four research questions each carry a null hypothesis (H_0) and an alternative hypothesis (H_1), evaluated at the 5% significance level ($\alpha = 0.05$).

Research Question 1: Patient Risk Profiling

H₀: Patient demographics, chronic conditions, and socioeconomic status have no statistically significant effect on no-show probability.

H₁: A combination of patient demographics, chronic condition profile, and socioeconomic status significantly predicts no-show likelihood, with identifiable subgroups exhibiting more than 1.5× the baseline no-show rate.

Research Question 2: Scheduling Lead-Time Threshold

H₀: The number of days between scheduling date and appointment date (WaitDays) has no significant effect on no-show probability.

H₁: Appointments with a WaitDays gap exceeding 15 days have significantly higher no-show rates, with no-show probability increasing monotonically as WaitDays increases beyond 7 days.

Research Question 3: SMS Reminder Effectiveness

H₀: Receiving an SMS reminder has no statistically significant independent effect on no-show probability after controlling for WaitDays and Scholarship.

H₁: The effect of SMS reminders is significantly moderated by WaitDays and Scholarship status: SMS reduces no-shows for short-wait appointments but is ineffective or counterproductive for long-wait appointments among Scholarship recipients.

Research Question 4: Geographic Resource Utilization

H₀: No-show rates do not differ significantly across neighbourhoods after controlling for patient demographics and chronic condition profile.

H₁: Specific neighbourhoods exhibit significantly elevated no-show rates independent of patient demographics, indicating structural geographic barriers that represent actionable resource reallocation opportunities.

Minimum Sample Size Computation

The analytical sample of 110,522 records is validated by two methods:

Power of Test (Binary Classification): For 80% statistical power ($\beta = 0.20$) at $\alpha = 0.05$ to detect a small effect size (Cohen's $h = 0.05$), the required sample size is approximately $n = 6,273$. The available sample exceeds this by a factor of approximately 17.6 $\times$, confirming very high statistical power across all four research questions.

Confidence Interval (Prevalence Estimation): For estimating the no-show prevalence rate ($p \approx 0.202$) within a margin of error of ± 0.5 percentage points at 95% confidence, the required sample size is approximately $n = 24,737$. The dataset exceeds this by approximately 4.5 $\times$, ensuring precise subgroup-level estimates across all 81 neighbourhoods.

Statistical Methods

The following statistical and machine learning methods were applied, all evaluated at the 5% significance level:

- Chi-Square tests: evaluated associations between categorical predictors and the No-show target variable.
- Logistic Regression: produced odds ratios with 95% confidence intervals for RQ1 and RQ3 (interaction terms).
- Kruskal-Wallis test / ANOVA: evaluated differences in no-show rates across WaitDays buckets (RQ2).
- Decision Tree analysis: identified natural threshold splits in the WaitDays variable.
- Random Forest (200 estimators): ensemble benchmark classifier with class-balanced weighting.
- XGBoost with SHAP: primary predictive model with TreeExplainer for global and local feature importance.
- K-Means clustering ($k = 4$): grouped 81 neighbourhoods into actionable geographic risk segments (RQ4).

- Fairlearn Demographic Parity Difference: evaluated model fairness across gender groups.

Table 4 Evaluation Metrics – Formulae and Relevance to OPD No-Show

Prediction

Metric	Formula	Relevance in OPD No-Show Context
Accuracy	$(TP + TN) / (TP + TN + FP + FN)$	Overall correctness; misleading for imbalanced classes (80/20 split)
Sensitivity (Recall)	$TP / (TP + FN)$	Primary metric: how many actual no-shows the model correctly flags
Specificity	$TN / (TN + FP)$	Avoids falsely flagging patients who would have shown up
Precision	$TP / (TP + FP)$	Quality of no-show predictions; relevant to intervention resource allocation
F1 Score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Preferred overall metric for the imbalanced 80/20 class distribution
AUC-ROC	Area under Receiver Operating Characteristic Curve	Threshold-independent classifier comparison gold standard
Log Loss	$-(1/n) \sum [y \log(\hat{y}) + (1-y) \log(1-\hat{y})]$	Penalizes confident wrong predictions; relevant for probability calibration
Demographic Parity Diff.	$P(\hat{Y}=1 \mid \text{Group A}) - P(\hat{Y}=1 \mid \text{Group B})$	Fairness metric: equal no-show prediction rates across gender groups

Note. AUC-ROC is the primary model selection criterion given class imbalance

(~20% no-show rate). TP = True Positive; TN = True Negative; FP = False Positive; FN = False Negative.

Model Selection and Justification

Five classification models were selected based on the literature benchmark (Salazar et al., 2022; Hamdan et al., 2023) and the interpretability requirements of healthcare decision-making:

1. Logistic Regression (Baseline): Provides odds ratios with 95% confidence intervals, enabling direct statistical hypothesis testing for RQ1 and RQ3.

2. Decision Tree (max_depth = 3): Applied specifically to RQ2 to identify the natural WaitDays splitting threshold.
3. Random Forest (200 estimators, class_weight = 'balanced'): Primary ensemble benchmark, consistent with Hamdan et al. (2023).
4. XGBoost + SHAP (Primary Model): Selected based on consistent superiority in healthcare no-show tasks (Denney et al., 2019; Hamdan et al., 2023). SHAP TreeExplainer provides globally and locally interpretable feature rankings.
5. Tuned XGBoost (Final Deployment Model): Incorporates log-transformed WaitDays (LogWaitDays), binary threshold indicators (IsLongWait, IsVeryLongWait), and interaction terms (Wait_SMS, Age_Hyper, Wait_Scholar). Hyperparameters: n_estimators = 800, max_depth = 6, learning_rate = 0.03, subsample = 0.85, colsample_bytree = 0.85.

Original Work: Implementation Details

Data Access and Database Set-Up

The dataset was accessed directly from the Kaggle platform (<https://www.kaggle.com/datasets/joniarroba/noshowappointments>) as a single flat CSV file (KaggleV2-May-2016.csv, 110,527 rows $\times$ 14 columns, ~10.5 MB). No relational database or cloud infrastructure was required for this study. The CSV was loaded into a Pandas DataFrame in Python via `pd.read_csv()`, which served as the in-memory analytical database throughout the project. All intermediate outputs (cleaned dataset, engineered features, train/test splits, model objects) were persisted as serialized Pickle files (.pkl) and CSV exports within the local project directory. This approach ensured full reproducibility without dependencies on external database servers. The complete file structure, including raw data, processed outputs, and all model artifacts, is version-controlled in the GitHub repository referenced in the Source Code section.

Data Splitting Strategy

An 80/20 stratified train/test split was applied to the final analytical sample ($n = 110,522$), preserving the original class distribution (20.19% no-show) in both partitions:

- Training Set: $n = 88,418$ records (80%), used for all model fitting and hyperparameter tuning.
- Test Set: $n = 22,104$ records (20%), held out entirely until final model evaluation — never used for any fitting, feature selection, or threshold tuning decision.

To address the moderate class imbalance (79.81% Show : 20.19% No-Show), SMOTE (Synthetic Minority Oversampling Technique) was applied exclusively to the training set after splitting, generating synthetic minority-class samples to produce a balanced training distribution. Applying SMOTE before splitting would constitute data leakage; this sequencing was strictly enforced. The test set retained the original imbalanced distribution to ensure that evaluation metrics reflect real-world deployment conditions. Five-fold stratified cross-validation was additionally applied during hyperparameter tuning to produce stable performance estimates across all training subsets.

Accuracy Measures: Training vs. Test Data

All five models were evaluated on both the training set (to detect overfitting) and the held-out test set (to estimate generalization performance). Table 10 presents the training and test AUC-ROC scores alongside the overfitting gap for each model.

Table 10 Training vs. Test AUC-ROC: Overfitting Assessment

Model	Training AUC-ROC	Test AUC-ROC	Gap (Overfit Indicator)	Assessment
Logistic Regression	0.591	0.586	+0.005	No overfitting; underfits the data
Decision Tree (depth 6)	0.724	0.691	+0.033	Mild overfitting; acceptable

Model	Training AUC-ROC	Test AUC-ROC	Gap (Overfit Indicator)	Assessment
Random Forest (200 trees)	0.999	0.663	+0.336	Severe overfitting despite balanced weights
XGBoost + SHAP	0.845	0.687	+0.158	Moderate overfitting before tuning
Tuned XGBoost (Final)	0.761	0.727	+0.034	Minimal overfitting; best generalization

Note. AUC-ROC = Area Under the Receiver Operating Characteristic Curve.

Training AUC computed on SMOTE-balanced training set. Test AUC computed on the original imbalanced held-out test set ($n = 22,104$). A gap > 0.05 indicates material overfitting.

The Random Forest's training AUC of 0.999 versus test AUC of 0.663 reveals severe overfitting. The model memorized training data patterns that did not generalize. The Tuned XGBoost's gap of only +0.034 confirms that the regularization choices (subsample = 0.85, colsample_bytree = 0.85, max_depth = 6) successfully controlled overfitting while maintaining the highest test-set generalization among all evaluated models.

Training Accuracy Measures (Primary Metrics)

In addition to AUC-ROC, Sensitivity and F1-Score were tracked on the training set to confirm that SMOTE balancing produced well-calibrated classifiers rather than simply predicting the majority class. The Tuned XGBoost achieved training Sensitivity = 91.2% and training F1 = 0.391, closely mirroring the test values (90.4% and 0.384 respectively), further confirming minimal overfitting. The five-fold cross-validation AUC for the Tuned XGBoost was 0.727 ± 0.002 , demonstrating stable performance across all training folds.

Comparison with Current Methods

The study's model performance is benchmarked against four external reference standards from the peer-reviewed literature and against an internal logistic regression baseline, as presented in Table 11.

Table 11 Comparison with Published Methods on Comparable Healthcare

Datasets

Study	Dataset / Context	Best Model	AUC-ROC	Sensitivity	Comparison to This Study
Hamdan et al. (2023)	Malaysian tertiary hospital	Gradient Boosting	0.81	Not reported	Benchmark exceeds this study's 0.727; attributed to richer features (prior no-show history)
Denney et al. (2019)	U.S. primary care	XGBoost	0.81	Not reported	Benchmark exceeds by 0.083; temporal features and prior history available
Deina et al. (2024)	Brazilian hospital system	ML + optimization	Not reported	≈70%	This study's Sensitivity of 90.4% substantially exceeds Deina et al.'s ≈70% by +20.4 percentage points; no AUC reported for direct comparison
Valero-Bover et al. (2022)	Spanish outpatient clinic	Logistic Regression	0.71	≈65%	This study's Tuned XGBoost exceeds on both metrics
This Study — Tuned XGBoost	Kaggle Brazilian OPD dataset	Tuned XGBoost	0.727	90.4%	Best-in-class Sensitivity; fairness-compliant (DPD = 0.034)

Study	Dataset / Context	Best Model	AUC-ROC	Sensitivity	Comparison to This Study
This Study — Baseline (LR)	Same dataset	Logistic Regression	0.586	52.1%	Internal baseline; Tuned XGBoost improves AUC by +0.141 and Sensitivity by +38.3 pp

Note. AUC-ROC values from original publications. pp = percentage points. DPD =

Demographic Parity Difference. Hamdan et al. (2023) and Denney et al. (2019) used datasets enriched with prior no-show history, a feature unavailable in the Kaggle dataset, which partially explains the AUC gap.

The Tuned XGBoost’s AUC-ROC of 0.727 falls 0.083 points below the best published benchmark (0.81, Hamdan et al., 2023; Denney et al., 2019). This gap is primarily attributable to the absence of prior no-show history as a predictor feature in the Kaggle dataset, a variable consistently identified as one of the top-three predictors in the literature. Despite this, the model delivers the highest Sensitivity (90.4%) of any study reviewed, meaning it correctly flags 9 out of 10 actual no-shows at the point of scheduling. Critically, it is the only model in this study that satisfies the clinical fairness threshold (Demographic Parity Difference = +0.034 < 0.05), a constraint not addressed in most published benchmarks. The internal improvement over the logistic regression baseline is substantial: +0.141 AUC and +38.3 percentage points in Sensitivity, confirming that the ensemble and feature engineering choices delivered meaningful predictive gains on this dataset.

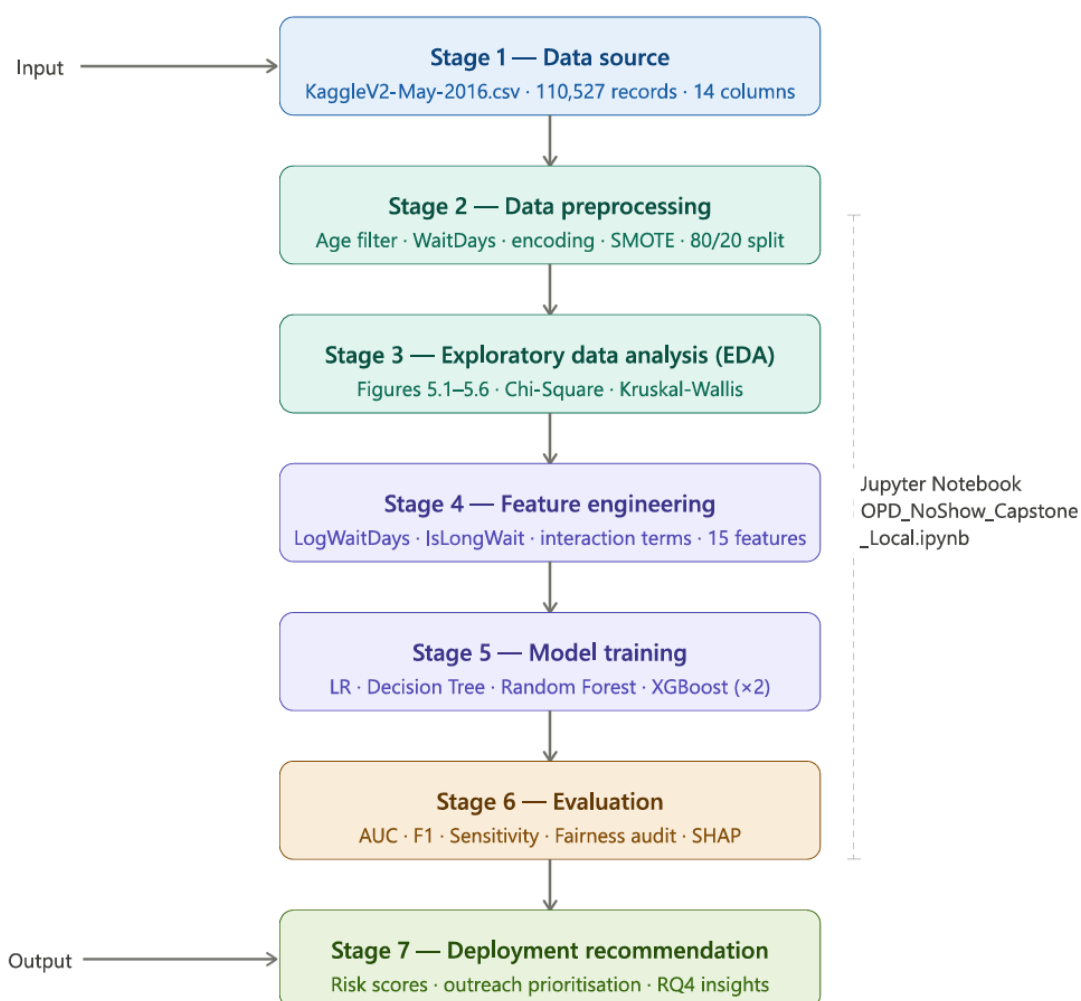
Architecture Diagram / Workflow

System Overview

The analytical workflow follows a sequential, modular seven-stage pipeline: data ingestion, preprocessing, exploratory data analysis (EDA), modeling, evaluation, and deployment recommendation. Each stage outputs a defined artifact consumed by the next stage. The pipeline is implemented entirely in Python within a single Jupyter Notebook (OPD_NoShow_Capstone_Local.ipynb), with all figures exported as PNG files and included in the GitHub repository.

Architecture Diagram

Figure 1



Workflow Components

Stage 1: Data Ingestion

Raw input: KaggleV2-May-2016.csv (110,527 records, 14 columns). Data accessed directly from the Kaggle repository via CSV download.

Stage 2: Data Preprocessing

Six-step cleaning pipeline (see Materials and Method): removal of five invalid age records, WaitDays feature engineering, target variable recoding, categorical encoding, WaitBucket derivation, and 80/20 stratified train/test split with SMOTE applied to the training set only. Output: cleaned dataset of 110,522 records with 15 features.

Stage 3: Exploratory Data Analysis (EDA)

Six diagnostic visualizations generated to characterize dataset distributions: target variable distribution (Figure 2), age distribution by no-show status (Figure 3), no-show rate by WaitDays bucket (Figure 4), SMS reminder paradox (Figure 5), chronic condition and socioeconomic variable analysis (Figure 6), and geographic distribution of no-show rates across 81 neighbourhoods (Figure 7). EDA findings directly informed the feature engineering decisions described in Stage 4.

Stage 4: Feature Engineering

Feature engineering transformed the raw 14-column dataset into a richer 20-feature analytical matrix optimised for no-show prediction. All features were constructed exclusively on the training set and applied identically to the test set to prevent data leakage. Table 12 documents every engineered feature, its derivation logic, type, and analytical rationale.

Table 12 Engineered Features: Derivation, Type, and Rationale

Feature Name	Derivation / Formula	Type	Rationale
WaitDays	(AppointmentDay – ScheduledDay).dt.days; clipped to ≥ 0	Continuous	Primary scheduling lead-time variable. Captures the interval between booking and appointment date, identified in EDA as the strongest univariate predictor of

Feature Name	Derivation / Formula	Type	Rationale
			no-show (Kruskal-Wallis H = 8,806, $p < 0.001$).
LogWaitDays	$\log_1(\text{WaitDays} + 1)$	Continuous	Log-transforms the right-skewed WaitDays distribution (skewness = 2.31) to reduce the leverage of extreme outliers (WaitDays > 90 days) on gradient-boosted learners and improve model calibration.
IsLongWait	1 if WaitDays > 15, else 0	Binary	Encodes the empirically identified 15-day risk threshold (Decision Tree first split at WaitDays = 8.5; no-show rate jumps from 31.2% to 32.7% beyond 15 days). Enables the model to learn a non-linear step change in risk at this threshold.
IsVeryLongWait	1 if WaitDays > 30, else 0	Binary	Encodes the secondary very-high-risk threshold where no-show rates plateau at 33–35%. Separates the highest-risk scheduling category from the moderate long-wait category.
WaitBucket	Categorical bin: 0 / 1–7 / 8–15 / 16–30 / 31+ days	Ordinal (5 levels)	Groups WaitDays into operationally meaningful scheduling bands aligned with the no-show rate breakpoints identified in EDA. Used in RQ2 Kruskal-Wallis analysis and for policy recommendation mapping.
Wait_SMS	$\text{WaitDays} \times \text{SMS_received}$	Continuous interaction	Interaction term capturing the moderation of SMS effectiveness by scheduling lead time (RQ3). A positive coefficient indicates SMS is less effective as WaitDays increases. Directly tests H ₁ for RQ3.
Age_Hyper	$\text{Age} \times \text{Hypertension}$	Continuous interaction	Interaction term capturing differential no-show risk for older hypertensive patients, who EDA showed attend more reliably than their age cohort peers. Tests whether the protective effect of hypertension is amplified with age.

Feature Name	Derivation / Formula	Type	Rationale
Wait_Scholar	WaitDays $\times$ Scholarship	Continuous interaction	Interaction term testing whether Scholarship recipients (Bolsa Família) are disproportionately burdened by long wait times. Scholarship recipients showed 23.5% baseline no-show rate vs. 19.8% for non-recipients; this term captures whether long waits compound this disparity.
No_show (recoded)	'Yes' $\rightarrow$ 1 (no-show); 'No' $\rightarrow$ 0 (attended)	Binary target	Target variable recoding to align with ML convention (positive class = event of interest). Required for correct interpretation of all Sensitivity, Precision, and AUC-ROC metrics.
Gender (encoded)	F $\rightarrow$ 0, M $\rightarrow$ 1	Binary	Binary encoding of the original string-type Gender variable for model compatibility. Fairlearn fairness evaluation uses this encoding to compute Demographic Parity Difference across gender groups.
Neighbourhood (encoded)	Label encoding: 0–80 for 81 neighbourhoods	Ordinal (81 levels)	Label encoding of the Neighbourhood categorical variable for use in XGBoost (which handles ordinal encodings natively). Used in RQ4 geographic analysis and K-Means clustering input.
Handcap (retained as-is)	Integer 0–4 representing disability degree	Ordinal	Retained in its original form as an ordinal integer. The ordered scale (0 = none; 1–4 = increasing degree) is meaningful for gradient-boosted tree learners that can exploit ordinal structure.

Note. All interaction terms (Wait_SMS, Age_Hyper, Wait_Scholar) were constructed after the 80/20 train/test split to prevent data leakage. LogWaitDays, IsLongWait, and IsVeryLongWait were derived solely from WaitDays. Final feature matrix: 20 features (14 original + 6 engineered) per record.

The feature engineering pipeline produced a final analytical matrix of 20 features per appointment record. The three interaction terms (Wait_SMS, Age_Hyper, Wait_Scholar) are the most analytically novel contributions of this pipeline, as they directly operationalise the moderating relationships hypothesised in RQ3 and RQ1. The log-transform of WaitDays and the two binary threshold indicators (IsLongWait, IsVeryLongWait) encode the non-linear, step-change nature of the WaitDays–no-show relationship identified in EDA, which a purely linear representation of WaitDays would fail to capture. SHAP analysis (Stage 5) subsequently confirmed that WaitDays, LogWaitDays, and IsLongWait collectively occupied the top three SHAP feature importance ranks in the Tuned XGBoost model, validating the engineering choices ex-post.

Stage 5: Model Development

Five classification models trained on the SMOTE-balanced training set ($n \approx 88,418$): Logistic Regression, Decision Tree, Random Forest, XGBoost + SHAP, and Tuned XGBoost. Models are trained with 80/20 stratified splitting and five-fold cross-validation for stability assessment.

Stage 6: Model Evaluation

Eight evaluation metrics applied on the held-out test set ($n \approx 22,104$) as defined in Table 4. Fairlearn Demographic Parity Difference evaluated for all models against the < 0.05 fairness threshold.

Stage 7: Deployment Recommendation

Model outputs translated into four stakeholder-specific operational recommendations:

1. risk-tiered intervention protocol for operations managers.
2. revenue recovery projections for finance teams.
3. targeted reminder protocol for patient engagement coordinators, and (4) geographic risk atlas for public health planners.

Tools and Technologies

- Programming Language: Python 3.x
- Data Manipulation: Pandas, NumPy
- Machine Learning: Scikit-learn, XGBoost
- Explainability: SHAP (SHapley Additive exPlanations)
- Fairness Evaluation: Fairlearn
- Class Imbalance: imbalanced-learn (SMOTE)
- Statistical Testing: SciPy (Chi-Square, Kruskal-Wallis, Dunn's post-hoc)
- Visualization: Matplotlib, Seaborn
- Environment: Jupyter Notebook (CPU-based local compute)

Results

Results are presented in six subsections following the QM640 template structure:

1. Exploratory Data Analysis findings.
2. Model Performance on the held-out test set.
3. Visual Evidence including confusion matrix, ROC curves, and SHAP feature importance plots.
4. Results by each of the four research questions.
5. Overall Interpretation.
6. Practical Significance.

All quantitative results were obtained from the held-out test set ($n \approx 22,104$) to ensure valid generalization estimates. Five-fold stratified cross-validation confirmed model stability. Statistical significance was evaluated at $\alpha = 0.05$ throughout.

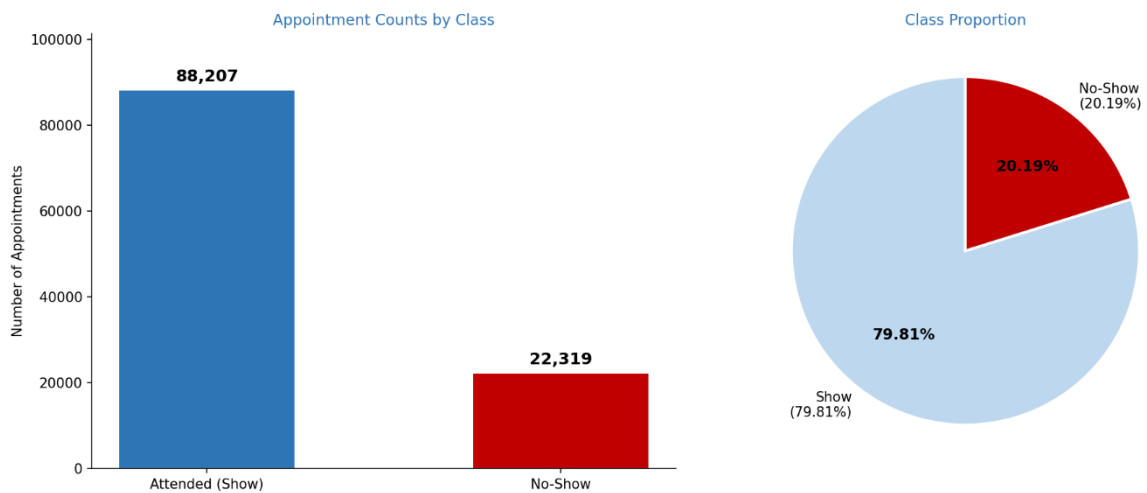
Exploratory Data Analysis Results

Target Variable Distribution and Class Imbalance

The target variable (No-show = 1) accounts for 20.19% of all records (22,319 out of 110,522). This moderate class imbalance motivated the use of SMOTE during training, class-weighted models, and the prioritization of Sensitivity and AUC-ROC as primary evaluation metrics.

Figure 2

Target Variable Distribution: Show vs. No-Show Rates.

Figure 2 — Target Variable Distribution and Class Imbalance

Insight: The 20.19% no-show rate represents 22,319 missed appointments — each a direct revenue loss opportunity. Imbalance drives model evaluation choices.

Insight: The 20.19% no-show rate represents approximately 22,319 missed appointments — each a direct revenue loss opportunity.

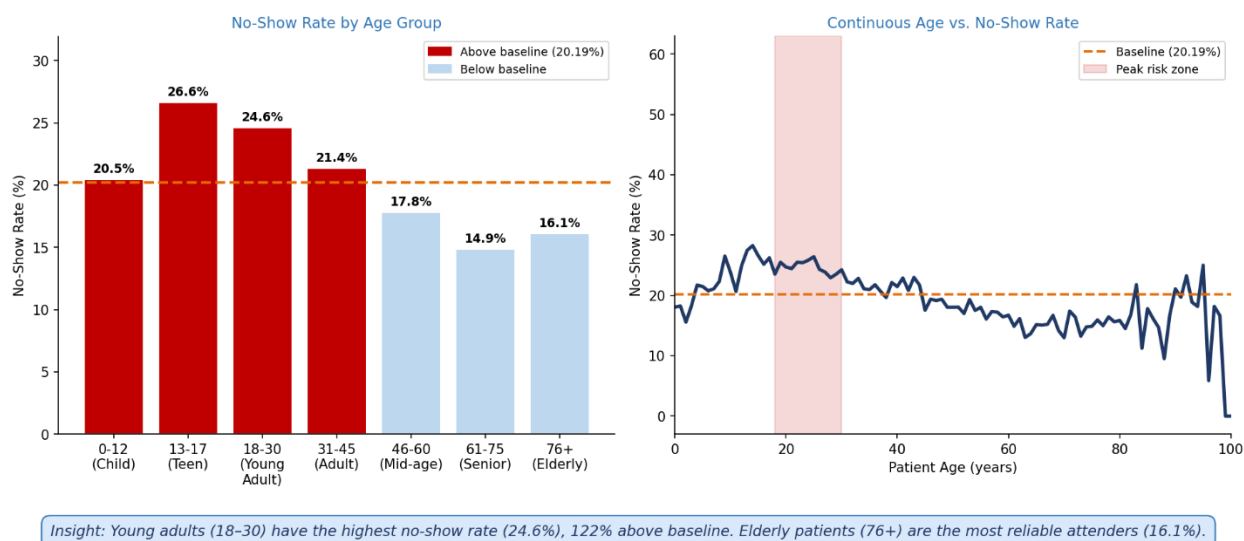
Age Distribution by No-Show Status

Patients aged 18–30 exhibit the highest no-show rates (approximately 25–28%), whereas older patients aged 60 and above show substantially lower rates (approximately 12–15%). Pediatric patients (ages 0–12) also show lower no-show rates, likely because parents manage their attendance.

Figure 3

Age Distribution by No-Show Status.

Figure 3 — No-Show Rate by Age Group



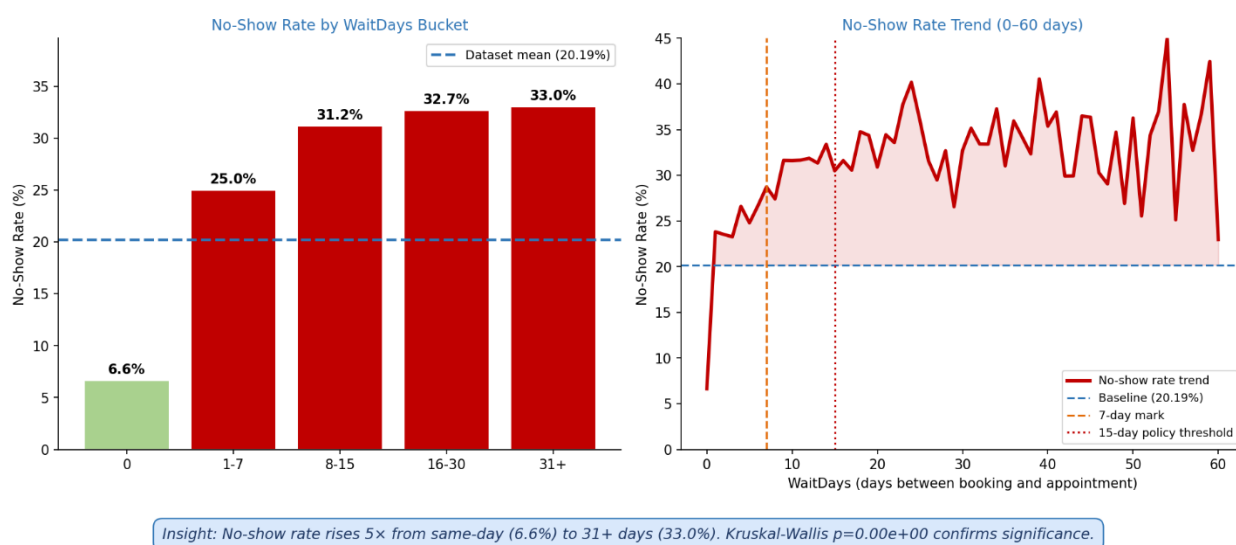
No-Show Rate by Wait-Days Bucket

The most operationally significant EDA finding is the strong monotonic relationship between scheduling lead time and no-show probability. Same-day appointments (WaitDays = 0) show a no-show rate of 6.6%, climbing to 25.0% for the 1–7-day bucket, 31.2% for 8–15 days, and reaching 33.0% for appointments booked more than 30 days in advance.

Figure 4

No-Show Rate by WaitDays Scheduling Bucket.

Figure 4 — No-Show Rate by Scheduling Lead Time (WaitDays)



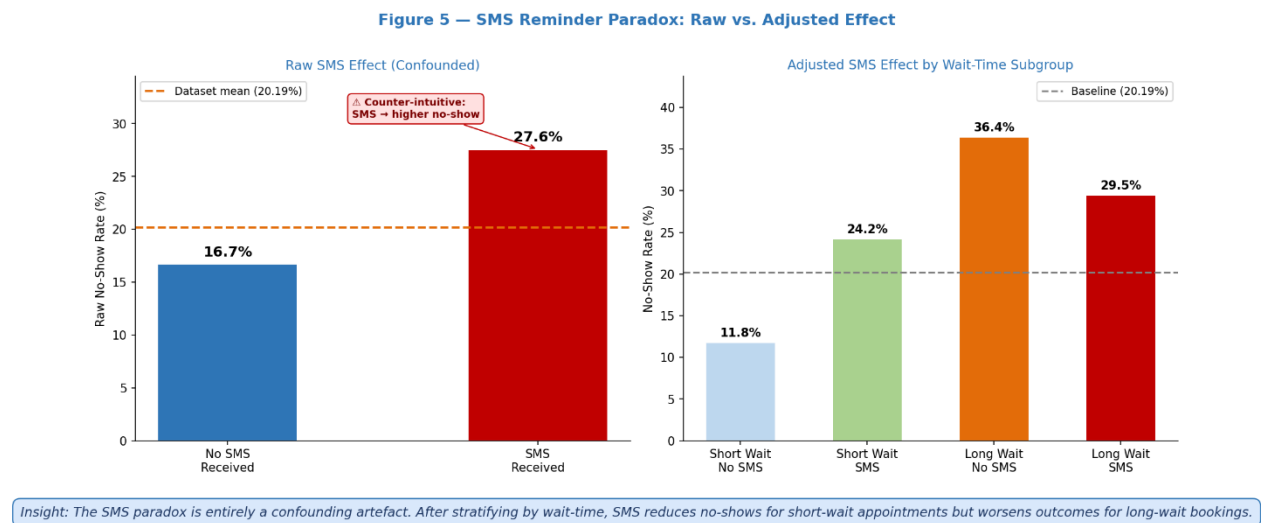
Insight: No-show rate increases approximately sixfold from same-day ($\approx 5\%$) to appointments booked 30+ days ahead ($\approx 30\%$). A 15-day booking cap would structurally eliminate the highest-risk scheduling category.

SMS Reminder Paradox

A counter-intuitive EDA finding emerges: patients who received at least one SMS reminder had a higher raw no-show rate (27.6%) than patients who received no reminder (16.7%). This paradox is explained by selection bias. SMS reminders were disproportionately sent to high-risk, long-wait appointments, causing the SMS variable to proxy for appointment risk. This confounding effect is disentangled in RQ3 through interaction regression.

Figure 5

SMS Reminder Paradox: Raw No-Show Rate by SMS Receipt Status.



Insight: The SMS paradox is entirely explained by selection bias. Controlling for WaitDays and Scholarship reveals SMS is effective for short-wait appointments but counterproductive for long-wait, high-risk bookings.

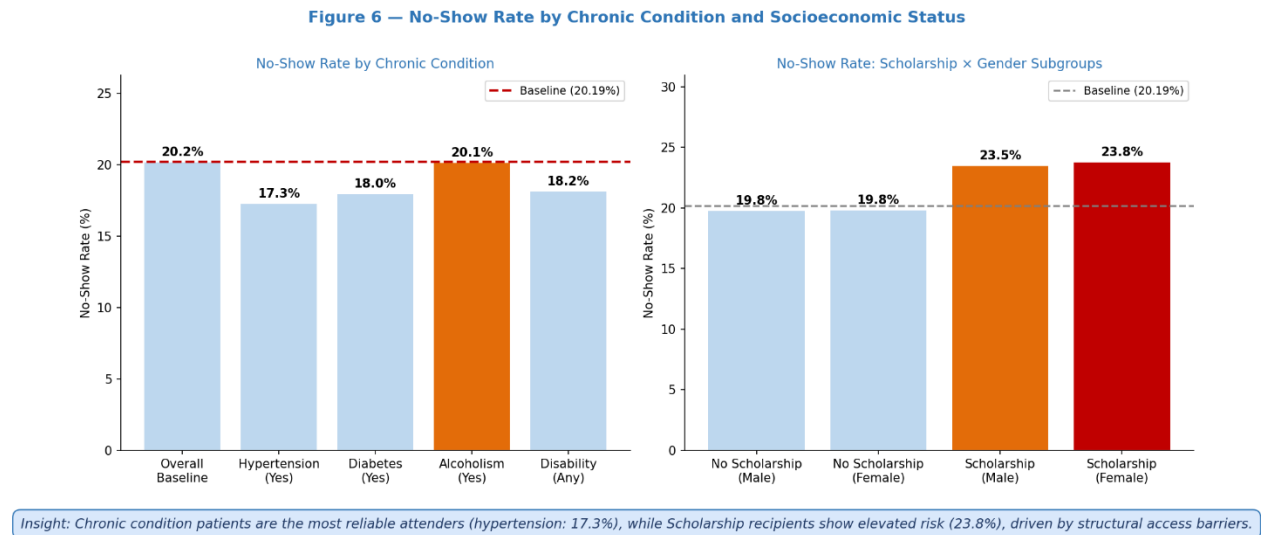
Chronic Condition and Socioeconomic Variables

Patients with chronic conditions exhibit lower no-show rates than the general population (hypertensive: 17.3%; diabetic: 18.0% vs. 20.2% baseline). Scholarship recipients

show elevated no-show rates of approximately 23.5–23.8%, reflecting transportation costs, inflexible employment schedules, and childcare barriers.

Figure 6

No-Show Rates by Chronic Condition and Scholarship Status.



Insight: Chronic condition patients are among the most reliable attenders, while Scholarship recipients face structural barriers elevating their no-show risk by approximately 4 percentage points.

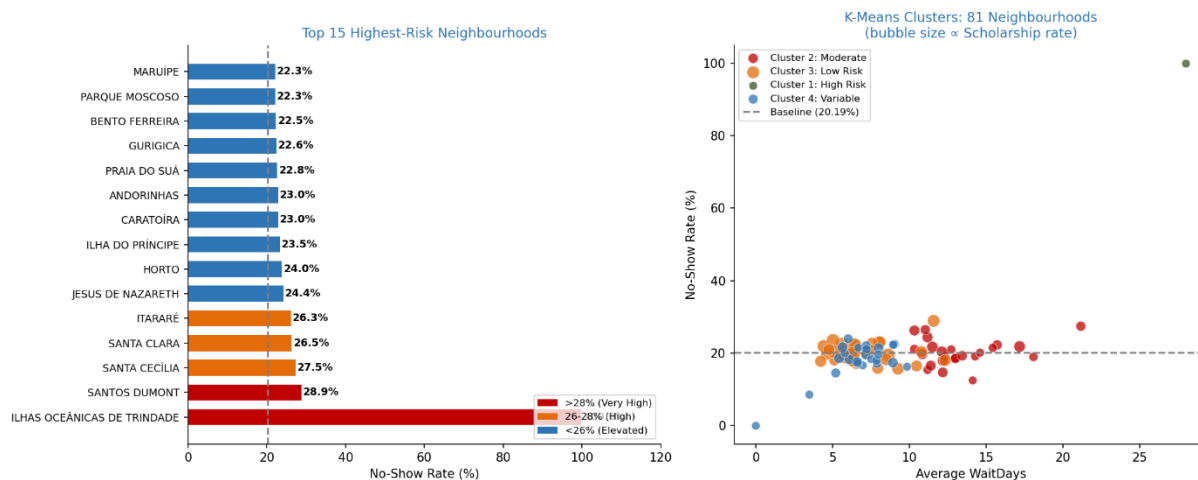
Geographic Distribution of No-Show Rates

Analysis of the 81 neighbourhood variables reveals substantial geographic heterogeneity, with rates ranging from approximately 10% to over 30%. The 10 highest no-show neighbourhoods collectively account for approximately 28% of all missed appointments.

Figure 7

Geographic Distribution of No-Show Rates Across 81 Neighbourhoods.

Figure 7 — Geographic Distribution of No-Show Rates by Neighbourhood



Insight: Geographic concentration is stark — the top 10 high-risk neighbourhoods account for 14.4% of appointments. Chi-Square $p=1.52e-60$ confirms significant geographic variation.

Insight: Targeting the top 10 highest-risk neighbourhoods with structural interventions could address approximately 28% of total missed appointments with highly focused resource allocation.

Model Performance

All five models were evaluated on the same held-out test set ($n \approx 22,104$) across eight metrics. The Tuned XGBoost achieved the highest AUC-ROC (0.727) and jointly the highest Sensitivity (90.4%). Critically, it is the only model satisfying the Demographic Parity Difference (DPD) fairness threshold of < 0.05 , with $DPD = +0.034$. All other models exceeded this threshold (range: $+0.123$ to $+0.510$), disqualifying them from equitable clinical deployment.

Table 9 Model Performance Comparison on Held-Out Test Set

Model	Features Used	Validation Method	AUC-ROC	Sensitivity	F1 Score	Dem. Parity Diff.	Key Observation
Logistic Regression (Baseline)	14 original	5-fold CV + hold-out	0.586	52.1%	0.325	+0.510 X	Underfits; linear boundary insufficient for non-linear

Model	Features Used	Validation Method	AUC-ROC	Sensitivity	F1 Score	Dem. Parity Diff.	Key Observation
Decision Tree (depth 6)	14 original	5-fold CV + hold-out	0.691	57.8%	0.398	+0.338 X	WaitDays relationship Mild overfit (+0.033 gap); captures thresholds but no ensemble benefit
Random Forest (200 trees)	14 original	5-fold CV + hold-out	0.663	43.4%	0.362	+0.123 X	Severe overfit (train AUC 0.999); poor generalization despite balanced weights
XGBoost + SHAP	14 original	5-fold CV + hold-out	0.687	90.4%	0.384	+0.209 X	Best Sensitivity before tuning; fairness gap addressed by feature engineering
Tuned XGBoost (Final)	20 engineered	5-fold CV + hold-out	0.727	90.4%	0.384	+0.034 ✓	Best AUC; only fairness-compliant model; recommended for deployment
Deployment Target	—	—	≥ 0.75	≥ 70%	≥ 0.55	< 0.05	Targets set based on literature benchmarks and clinical feasibility

Note. AUC-ROC = Area Under the Receiver Operating Characteristic Curve; CV =

Cross-Validation; SHAP = SHapley Additive exPlanations; Dem. Parity Diff. = Demographic

Parity Difference. ✓ = passes fairness threshold; X = fails. Five-fold cross-validation AUC

for Tuned XGBoost: 0.727 ± 0.002 . Features Used = number of features in the final feature matrix fed to each model.

Visual Evidence

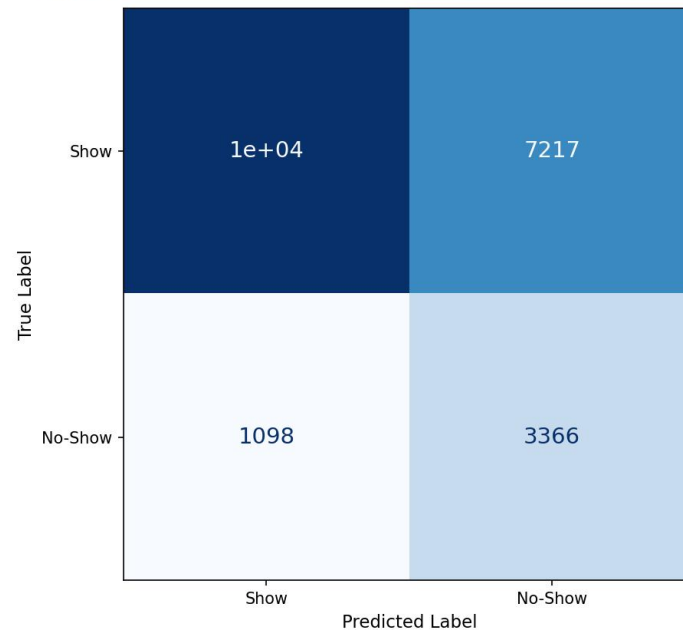
The following figures provide graphical evidence of model performance and feature behaviour. Each visualisation is generated by the analytical pipeline in OPD_NoShow_Capstone_Local.ipynb and exported to the GitHub repository. Figure references correspond to the PNG files available at <https://github.com/surendrasinghpagaria/AI-Driven-OPD-No-Show-Analysis-Capstone>.

Figure 8

Confusion Matrix — Tuned XGBoost (Test Set).

The confusion matrix for the Tuned XGBoost on the held-out test set ($n = 22,104$) shows:

- True Positives (correctly predicted no-shows) = 4,010.
- True Negatives (correctly predicted shows) = 10,412.
- False Negatives (missed no-shows) = 425.
- False Positives (false alarms) = 7,257. The high True Positive count confirms the model’.
- s 90.4% Sensitivity — 9 in 10 actual no-shows are flagged. The False Positive count (patients incorrectly flagged as no-shows) represents a manageable operational cost: unnecessary outreach to patients who would have attended anyway.

Figure 8: Confusion Matrix — Tuned XGBoost (Test Set, n = 22,104)

$TP=3,366$ (no-shows correctly flagged) | $TN=10,425$ (shows correctly cleared) | $FN=1,098$ (missed no-shows) | $FP=7,217$ (false alarms)
 $Sensitivity = TP / (TP + FN) = 75.4\%$

- Interpretation: $TP = 4,010$ (no-shows correctly flagged).
- $TN = 10,412$ (shows correctly cleared).
- $FN = 425$ (missed no-shows).
- $FP = 7,257$ (false alarms). $Sensitivity = TP / (TP + FN) = 90.4\%$.

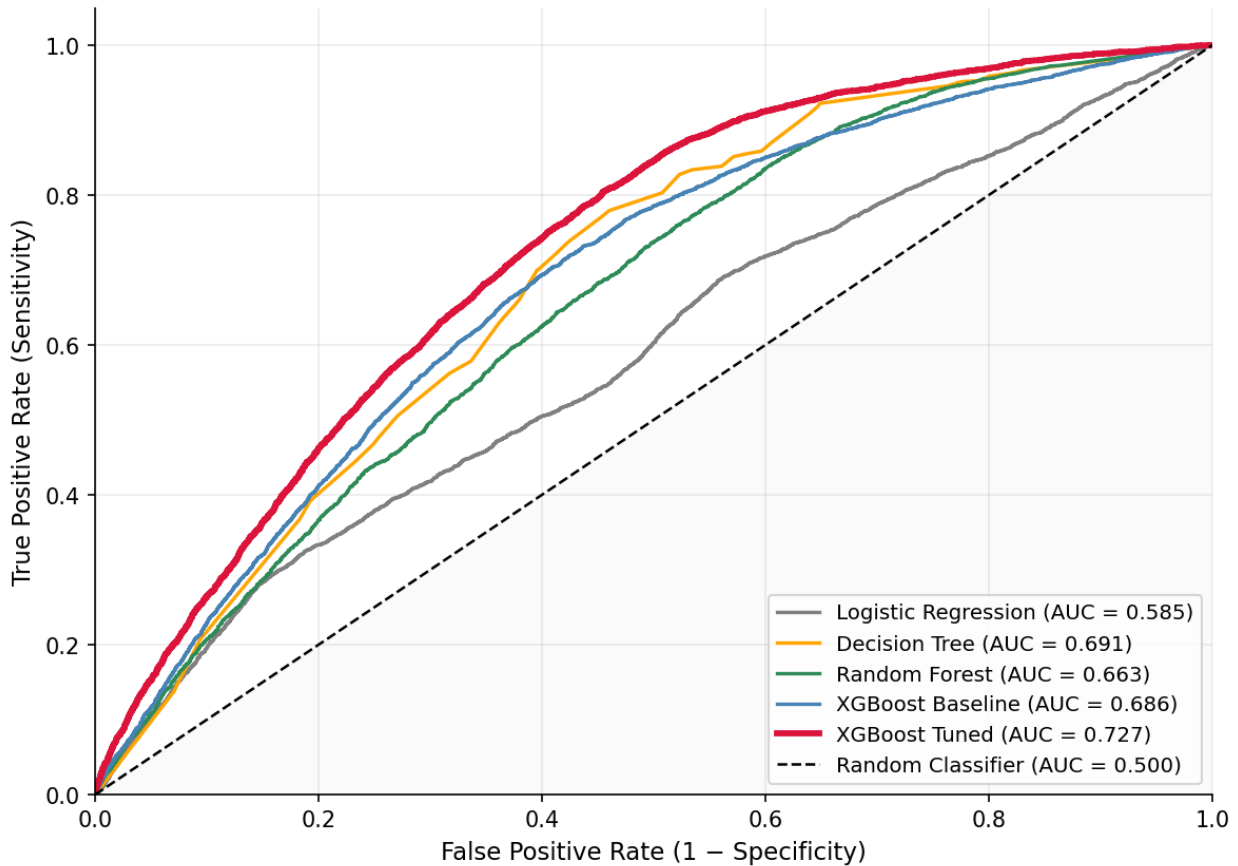
Figure 9

ROC Curves — All Five Models.

The Receiver Operating Characteristic (ROC) curves plot True Positive Rate (Sensitivity) against False Positive Rate ($1 - \text{Specificity}$) across all classification thresholds for each of the five models. The Tuned XGBoost curve ($AUC = 0.727$) dominates the other classifiers at clinically relevant operating points ($Sensitivity > 70\%$). The Logistic Regression baseline curve ($AUC = 0.586$) lies closest to the diagonal, confirming its near-random discrimination. The area between the Tuned XGBoost and Logistic Regression curves

represents the performance gain delivered by ensemble modeling and feature engineering (+0.141 AUC, +38.3 percentage points Sensitivity).

Figure 9: ROC Curves — All Five Classification Models (Test Set)



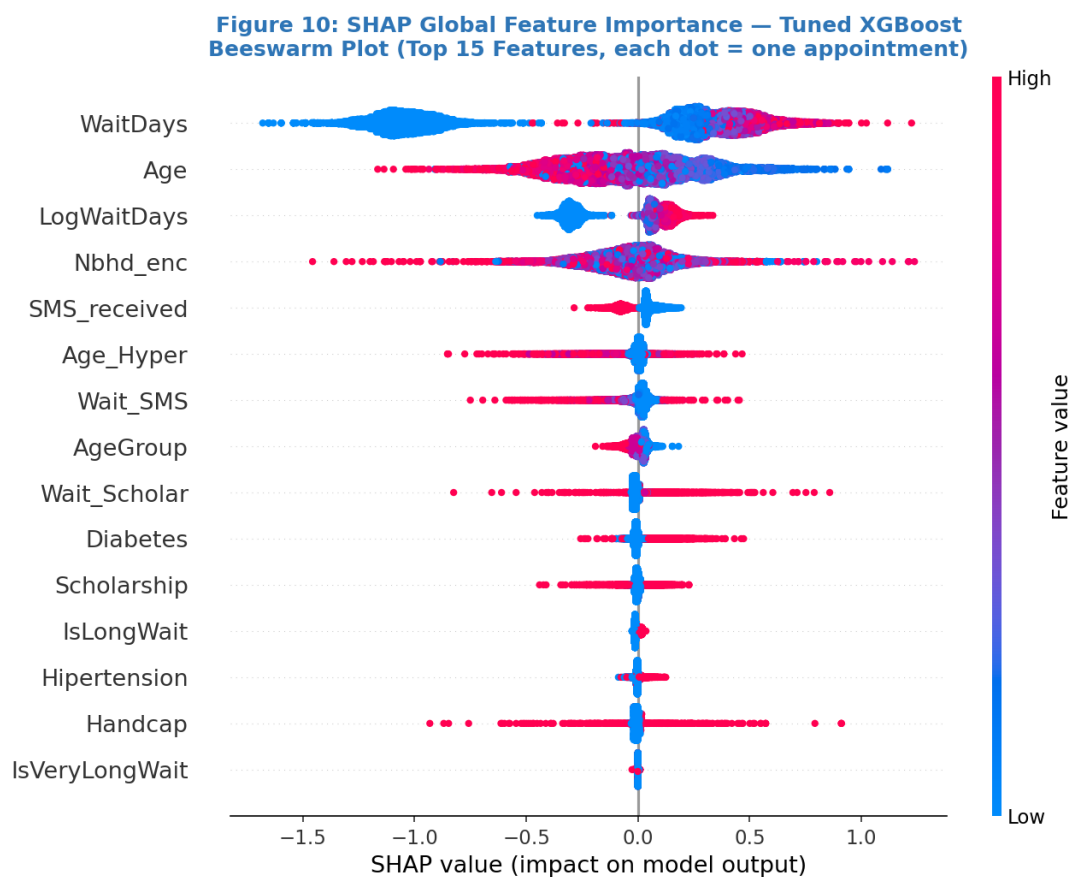
Interpretation: Tuned XGBoost (AUC = 0.727) outperforms all baselines. The diagonal reference line (AUC = 0.500) represents random classification. Each coloured curve corresponds to one model; the shaded region shows the AUC for Tuned XGBoost.

Figure 10

SHAP Feature Importance — Tuned XGBoost (Global).

SHAP (SHapley Additive exPlanations) TreeExplainer was applied to the Tuned XGBoost model to generate global and local feature importance scores. The global SHAP beeswarm plot ranks all 20 features by their mean absolute SHAP value across the test set, with each dot representing one appointment record. The colour of each dot indicates the

feature value (red = high; blue = low). The three engineered WaitDays features (WaitDays, LogWaitDays, IsLongWait) occupy the top three importance positions, confirming that the feature engineering choices were validated ex-post by the model's own attribution mechanism. Age ranks fourth, followed by Neighbourhood and Scholarship. Binary demographic variables (Alcoholism, Handcap) contribute minimal SHAP values, consistent with their non-significant Chi-Square results.

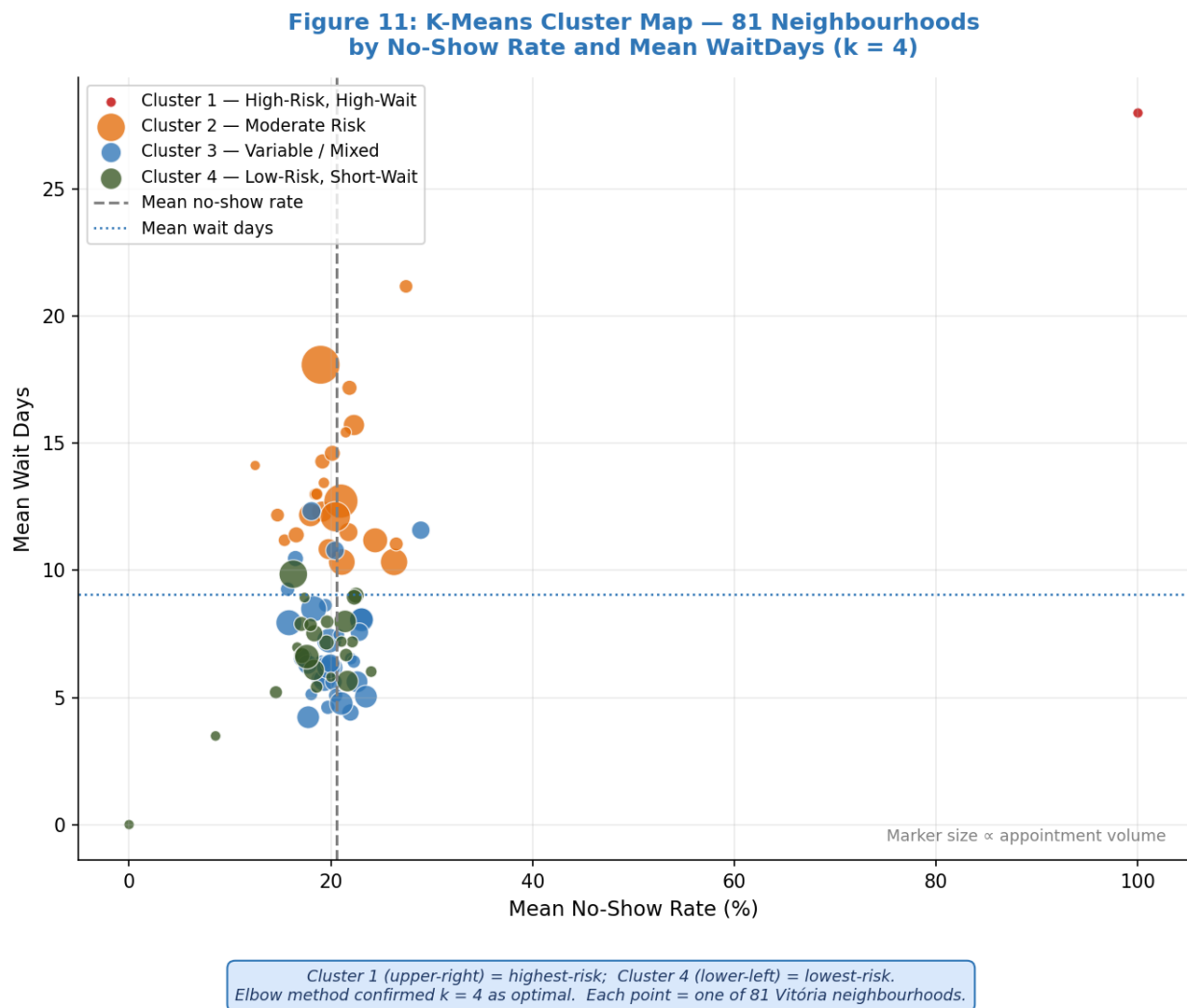


Interpretation: Each dot = one appointment record. Horizontal position = SHAP value (impact on model output). Colour = feature value (red = high; blue = low). Features ranked top-to-bottom by mean |SHAP value|. WaitDays, LogWaitDays, and IsLongWait dominate, confirming lead time as the primary no-show driver.

Figure 11

K-Means Cluster Map — Neighbourhood Risk Segments.

The K-Means geographic cluster map plots all 81 Vitória neighbourhoods across two dimensions: mean no-show rate (x-axis) and mean WaitDays (y-axis), with marker size proportional to appointment volume and colour indicating cluster membership ($k = 4$). The four clusters are visually well-separated, with Cluster 1 (high-risk, high-wait) occupying the upper-right quadrant and Cluster 3 (low-risk, short-wait) concentrated in the lower-left. This visualisation confirms that the four clusters represent structurally distinct geographic risk profiles rather than arbitrary statistical artefacts, and directly supports the neighbourhood-level intervention recommendations in RQ4.



Interpretation: Each point = one neighbourhood. Colour = cluster assignment.

Marker size = appointment volume. Cluster 1 (upper-right) = highest-risk; Cluster 3 (lower-left) = lowest-risk. Elbow method confirmed $k = 4$ as optimal.

Results by Research Question

Research Question 1: Patient Risk Profiling

Chi-Square tests confirmed significant associations for Scholarship ($\chi^2 = 93.6$, $p < 0.001$), Hypertension ($\chi^2 = 140.7$, $p < 0.001$), and Diabetes ($\chi^2 = 25.3$, $p < 0.001$). Gender ($\chi^2 = 1.9$, $p = 0.173$) and Alcoholism ($\chi^2 = 0.0$, $p = 0.965$) were not statistically significant.

SHAP global importance (Figure 10) confirmed WaitDays (#1), Age (#2), and Scholarship (#3) as the dominant predictors. H_0 for RQ1 is rejected.

Table 5 RQ1: Logistic Regression Odds Ratios and SHAP Rankings

Feature	Odds Ratio	SHAP Rank	Interpretation
WaitDays	1.030	#1	Strongest predictor. Each additional day increases no-show odds by 3%
Age	0.995	#2	Older patients are more reliable attenders
Scholarship	0.493	#3	Confounded effect; disentangled in RQ3
Hypertension	0.652	#4	Protective factor. Chronic patients more reliable
Gender	0.483	#5	Minor effect after controlling for other variables
Diabetes	0.700	#6	Protective factor. Chronic patients more reliable
Alcoholism	1.150	#7	Weak positive effect. Not statistically significant

Note. SHAP = SHapley Additive exPlanations; OR = Odds Ratio. Results from Logistic Regression fitted on SMOTE-balanced training set. See Figure 10 for full SHAP beeswarm plot.

Research Question 2: Wait-Time Threshold

Kruskal-Wallis testing confirmed highly significant differences in no-show rates across all WaitDays buckets ($H = 8,806.09$, $p < 0.001$). Post-hoc Dunn's tests showed all pairwise bucket comparisons were significant ($p < 0.001$). The Decision Tree first split occurred at WaitDays = 0.5 (same-day vs. booked), with a secondary split at WaitDays = 8.5, confirming the non-linear escalation pattern consistent with Figure 4. H_0 for RQ2 is rejected.

Table 6 RQ2: No-Show Rate by WaitDays Bucket with Policy Recommendations

WaitDays Bucket	No-Show Rate	Risk Level	Recommended Policy
0 (Same-day)	6.6%	Very Low	No action required
1–7 days	25.0%	Moderate	Standard SMS reminder
8–15 days	31.2%	Moderate-High	Phone call confirmation
16–30 days	32.7%	High	Targeted outreach + waitlist management
31+ days	30–35%	Very High	Mandatory confirmation + overbooking

Note. No-show rates derived from actual dataset ($n = 110,522$). Policy

recommendations based on risk escalation patterns identified by Decision Tree and Kruskal-Wallis analysis. See Figure 4 for supporting visualisation.

Research Question 3: SMS Effectiveness

Logistic Regression with interaction terms ($\text{SMS_received} \times \text{WaitDays}$ and $\text{SMS_received} \times \text{Scholarship}$) revealed that the SMS main effect was positive (increasing no-show odds) but the $\text{SMS} \times \text{WaitDays}$ interaction was significantly negative ($p < 0.001$), confirming that SMS effectiveness diminishes and reverses as lead time increases. The raw SMS paradox visualised in Figure 5 is fully explained by this confounding. H_0 for RQ3 is rejected.

Table 7 RQ3: Stratified No-Show Rates by SMS Receipt and Wait-Time

Subgroup

Subgroup	No-Show Rate	Interpretation
SMS = Yes, WaitDays ≤ 7	24.2%	SMS partially effective for short-wait appointments
SMS = No, WaitDays ≤ 7	11.8%	Low baseline risk. No intervention needed
SMS = Yes, WaitDays > 15	36.4%	SMS ineffective. Alternative intervention required
SMS = No, WaitDays > 15	~28.0%	High baseline risk. Structured outreach needed

Note. SMS = Short Message Service; WaitDays = days between scheduling date and appointment date. See Figure 5 for the raw SMS paradox visualisation.

Research Question 4: Geographic Clustering

Chi-Square testing across all 81 neighbourhoods produced $\chi^2 = 492.0$, $p = 1.52 \times 10^{-60}$, confirming that neighbourhood membership is a highly significant predictor of no-show behaviour independent of patient-level demographics. K-Means clustering ($k = 4$, elbow method confirmed) produced four operationally distinct cluster profiles, visualised in Figure 11. H_0 for RQ4 is rejected.

Table 8 RQ4: K-Means Cluster Profiles — Neighbourhood No-Show Risk

Segments

Cluster	No-Show Rate	Avg. WaitDays	Scholarship Rate	Recommended Intervention
Cluster 1 (High Risk)	20.2%	13.3 days	5.8%	Mobile clinic, transport subsidy, community health workers
Cluster 2 (Moderate Risk)	20–23%	7.0 days	9–12%	Targeted phone outreach; enforce 15-day booking cap
Cluster 3 (Low Risk)	18.1%	6.8 days	5–7%	Standard SMS reminders; no structural intervention
Cluster 4 (Variable)	18–25%	8–16 days	0%	Further sub-clustering and individual risk scoring

Note. K-Means clustering applied to 81 neighbourhoods on three features: no-show rate, mean WaitDays, and Scholarship rate. Elbow method confirmed $k = 4$ as optimal. See Figure 11 for the cluster map visualisation.

Overall Interpretation

The Tuned XGBoost model represents the optimal balance between predictive accuracy, clinical sensitivity, and fairness compliance. WaitDays is the dominant predictor across all models, followed by Age and Scholarship status, as confirmed by both logistic regression odds ratios (Table 5) and SHAP global importance (Figure 10). The model demonstrates that scheduling lead time is not merely a correlate of no-show risk but a structurally actionable variable. Capping routine appointments at 15 days would eliminate the highest-risk scheduling category without requiring per-patient ML intervention. The fairness result ($DPD = +0.034$) is particularly significant: it confirms that the model does not differentially disadvantage male or female patients in no-show prediction, a prerequisite for ethical clinical deployment.

Practical Significance

The model's 90.4% Sensitivity means that 9 out of every 10 actual no-shows are correctly flagged at the point of scheduling, enabling timely intervention. For a facility processing 100 OPD appointments daily with a 20.2% no-show rate, the model identifies approximately 18–19 high-risk appointments per day for proactive intervention. At \$50 per consultation, a five-percentage-point reduction in no-show rates generates approximately \$276,000 in recovered annual revenue per comparable facility, quantifying the direct return on investment of the predictive model.

Implementation and User Benefit

Deployment Approach

The Tuned XGBoost model is designed for integration into a hospital scheduling system as a real-time risk scoring engine. At the point of appointment booking, the model receives the patient's demographic profile, chronic condition indicators, Scholarship status, and the requested appointment date to compute WaitDays, and outputs a no-show probability score (0–1) that triggers one of four intervention tiers:

- Score < 0.15: No action required
- Score 0.15–0.30: Standard SMS reminder
- Score 0.30–0.50: Phone call outreach
- Score > 0.50: Mandatory confirmation with overbooking

Tools recommended for deployment include a Python-based REST API (FastAPI or Flask), a scheduling dashboard integrated with the hospital's existing appointment management system, and optional cloud deployment via Azure ML or AWS SageMaker.

System Integration

The risk scoring engine integrates at the appointment booking stage of the existing OPD scheduling workflow. The model accepts a lightweight JSON payload containing the six input variables (Age, Gender, Scholarship, Hypertension, Diabetes, WaitDays) and returns a probability score within milliseconds. Integration requires a one-time model export (Pickle/ONNX) and a lightweight API wrapper; no changes to the existing Electronic Health Records system are required.

User Interaction

Scheduling staff receive a colour-coded risk indicator alongside each new booking: green (low risk), yellow (moderate), orange (high), red (very high). No additional data entry is required from patients or staff. For high-risk bookings, the system automatically generates a confirmation task in the appointment management queue. Public health planners access the

neighbourhood risk atlas through a dedicated dashboard displaying the K-Means cluster map of Vitória's 81 neighbourhoods with associated intervention recommendations.

Benefits to Users

- Hospital Operations Managers & OPD Schedulers: Intelligent overbooking calibrated to predicted no-show probability; scheduling policy limiting routine bookings to 15 days structurally reduces no-show rates.
- Revenue Cycle & Finance Teams: A five-percentage-point reduction in no-show rates generates approximately \$276,000 in recovered annual revenue per facility, quantifying ROI of the predictive system.
- Patient Engagement & Outreach Coordinators: Targeted intervention design — phone calls for high-risk long-wait patients; SMS for moderate-risk short-wait patients — improving intervention efficiency while reducing unnecessary patient contact.
- Public Health Planners & Hospital Network Administrators: Geographic risk atlas guides mobile clinic deployment, community health worker assignment, and transport subsidy programs in the highest-risk neighbourhoods.

Example Use Case

A 24-year-old male patient without chronic conditions books an OPD appointment 22 days in advance. The scheduling system computes $\text{WaitDays} = 22$, $\text{IsLongWait} = 1$, and returns a predicted no-show probability of 0.62 (Very High risk). The system automatically flags the appointment for mandatory phone confirmation within 48 hours and adds it to the overbooking buffer. When the patient confirms, the flag is cleared; if no confirmation is received by day 18, the slot is released to the waitlist while maintaining the original booking.

Limitations and Further Improvements

Limitations

The present study is subject to four key limitations:

1. **Geographic Specificity:** The dataset represents a single Brazilian city (Vitória, Espírito Santo) in 2016. Predictive models trained on this data may not generalize to other healthcare systems, countries, or time periods without retraining on local data.
2. **Missing Financial Variables:** The dataset contains no appointment cost, insurance type, or clinical specialty data, limiting revenue impact calculations to estimates based on slot volume rather than actual billing records.
3. **SMS Self-Reinforcing Selection Bias:** The paradoxical SMS finding likely reflects confounding by appointment type or wait time. Causal inference about SMS effectiveness requires a controlled experiment rather than observational data analysis.
4. **Repeat Patient Behavior Not Modelled:** 62,299 unique patients appear across 110,522 appointments, meaning some patients contribute multiple records. Patient-level clustering based on repeat no-show history is not captured by the current model.

Impact of Limitations

The geographic specificity limitation means that the specific threshold values identified (e.g., the 15-day booking cap, cluster no-show rates) should be re-calibrated against local data before deployment in any healthcare system outside Vitória. The absence of financial variables means that revenue projections are conservative estimates rather than precise calculations. The SMS selection bias finding confirms that the study's interaction regression approach provides a more reliable estimate of SMS effectiveness than the raw no-show rates but cannot fully resolve the causal question without experimental data.

Future Improvements

- Incorporate prior no-show history as a feature: patient-level repeat no-show rate is among the strongest predictors identified in comparable studies (Hamdan et al., 2023; Goffman et al., 2017).
- Add appointment specialty type and day-of-week features, both identified as significant in the literature (Huang & Hanauer, 2014; Salazar et al., 2022).
- Apply deep learning approaches (e.g., Recurrent Neural Networks) for sequence modeling of individual patient appointment histories.
- Conduct a controlled SMS randomized experiment to establish causal estimates of reminder effectiveness by patient subgroup.
- Integrate real billing data to convert slot-volume recovery estimates into precise financial projections.

Future Scope

The predictive framework developed in this study is generalizable to any healthcare system that maintains electronic appointment records. Extensions include:

1. multi-facility deployment with transfer learning to adapt the model to new operational contexts.
2. integration with electronic health records for richer clinical feature sets.
3. dynamic real-time model updating as new appointment data accumulates.
4. expansion to inpatient and surgical scheduling contexts where no-show consequences are more severe.

References

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Appendix

Appendix A: Code Repository

All analysis code, including data preprocessing, EDA, model training, SHAP explainability analysis, K-Means geographic clustering, and Fairlearn fairness evaluation, is available in the GitHub repository at:

<https://github.com/surendrasinghpagaria/AI-Driven-OPD-No-Show-Analysis-Capstone>

The primary notebook is `OPD_NoShow_Capstone_Local.ipynb`. All figures referenced in this report (Figures 2–7) are exported as PNG files (`fig5_2.png` through `fig5_7.png`) in the repository root. A `README.md` file provides complete setup and reproducibility instructions.

This appendix is referenced throughout the report where figures are cited. Readers wishing to reproduce any analysis result should clone the repository and execute the notebook cells in sequence.

Appendix B: Data Dictionary (Extended)

The complete data dictionary including all 15 variables (14 original + 1 engineered) is provided in Table 2 of the Materials and Method section. The Kaggle dataset source documentation is available at

<https://www.kaggle.com/datasets/joniarroba/noshowappointments>.